

The Three Horsemen of Riches: Plague, War, and Urbanization in Early Modern Europe

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Introduction: The European Puzzle

- **The Puzzle:** In 14th-century Europe, nothing suggested future riches. Literacy was low, politics fragmented, and China appeared far more promising.
- **The Reality:** By 1700, Western European p.c. income grew by 30%, and urbanization rates tripled.
- **The Question:** How did Europe escape the “Iron Law of Wages”?
- **Standard View:** Productivity gains are usually swallowed by population growth (Malthusian Trap).
- **The Authors’ Thesis:** A unique **mortality regime** (the “Horsemen Effect”) turned temporary wage gains into permanent riches.

The First Divergence (1350–1700)

- Long before the Industrial Revolution, Europe pulled ahead of China and the Middle East.
- **Urbanization as a Proxy:**
 - Europe: 3% (pre-plague) → 9% (1700).
 - China: Stagnated at 3–4% for centuries.
- **The Paradox:** This “Rise of Europe” happened despite (and because of) extreme political fragmentation and frequent deadly disease outbreaks.
- **Empirical Fact:** Wage gains after the Black Death were not fully reversed in Europe, unlike in other Malthusian settings.

Figure 1: Conceptual Framework

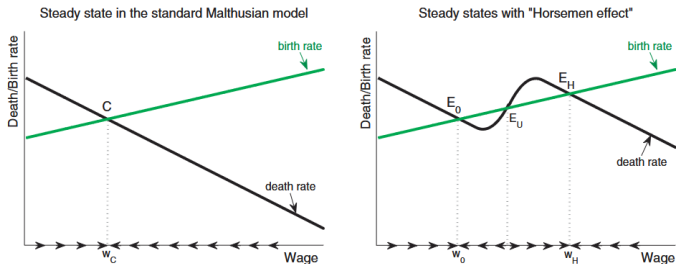


FIGURE 1

Steady states in the standard Malthusian model and with "Horsemen effect"

- **Multiple Steady States:**
- E_0 : Low mortality, low income (China).
- E_H : High mortality, high income (Europe).
- **Ratchet Effect:** A major shock (Black Death) pushes the system from E_0 past E_U to E_H .

Mechanism I: Unhealthy Cities (The First Horseman)

- **Cities as “Graveyards”:** Early modern cities were incredibly unhealthy. Urban mortality was often 1.8x higher than in the countryside.
- **The Rural-Urban Differential:** Life expectancy was 50% higher in the countryside.
- **Engel’s Law:** As survivors of the Black Death became richer, they spent more on urban manufactured goods.
- **Result:** This shift in demand led to larger (but deadlier) cities, which helped check population growth and sustain high wages.

Mechanism II: War and Taxes (The Second Horseman)

- **War as a Luxury Good:** Fragmented European states spent up to 80% of their revenue on warfare.
- **The Income-War Link:** Higher p.c. incomes → Higher tax revenues → More frequent and longer-lasting wars.
- **The Indirect Killers:** Armies rarely killed many on the battlefield. Instead, they moved through remote areas, spreading diseases (plague, typhus, smallpox) to unexposed populations.
- **Impact:** War acted as a “neutron bomb,” raising mortality without destroying much productive capital.

Mechanism III: Trade (The Third Horseman)

- **Economic Integration:** Higher incomes stimulated trade in bulky goods.
- **Disease Vectors:** Trade routes served as the primary highways for the plague and other epidemics.
- **Interaction Effects:**
 - War disrupted administrative procedures (like quarantines).
 - The combination of War, Trade, and Cities created a self-reinforcing cycle of high mortality and high income.

The Model: Consumption and Demand

- **Non-homothetic Preferences (Stone-Geary):**

$$u(c_A, c_M) = \begin{cases} (c_A - \underline{c})^\alpha c_M^{1-\alpha} & \text{if } c_A > \underline{c} \\ \varphi(c_A - \underline{c}) & \text{if } c_A \leq \underline{c} \end{cases}$$

- $\underline{c}$ is the subsistence food level. Consumers must satisfy food needs before buying urban goods.
- **Engel curve for Urbanization:**

$$n_M = (1 - \alpha) \left(1 - \frac{\underline{c}}{w}\right)$$

- As wage w increases, the share of labor in manufacturing (n_M) and the urbanization rate grow.

The Model: Production and Equilibrium

- **Agriculture (Rural):** Uses labor N_A and fixed land L .

$$Y_A = A_A N_A^\beta L^{1-\beta} \implies w_A = A_A (l/n_A)^{1-\beta}$$

- **Manufacturing (Urban):** Uses labor N_M only.

$$Y_M = A_M N_M \implies w_M = p_M (1 - \tau) A_M$$

- **Labor Mobility:** $w_A = w_M = w$.
- **Key Dynamic:** Population growth depends on w , which in turn depends on population size (decreasing returns in agriculture).

The Horsemen Function $h(n_M)$

- **Aggregate Death Rate (d):**

$$d = [1 + h_W(n_M) + h_M(n_M)] \cdot \bar{d}(c_A/\underline{c})^{\varphi_d}$$

- $h_M(n_M)$: Direct urbanization effect (city penalty).
- $h_W(n_M)$: War and trade effect (S-shaped).
- **The Threshold Effect:** Warfare only spreads disease once tax revenues cover fixed costs f_W . This creates the unstable middle steady state E_U .

Table 1: Baseline Calibration (Full)

Symbol	Interpretation	Value
α	Responsiveness of urbanization to income	0.68
β	Labour share in agriculture	0.6
$\underline{c}$	Subsistence food consumption	1
L	Land	1
$A_{A,0}, A_{M,0}$	Initial TFP (Agri, Mfg)	1.076, 2.265
γ_A	Rate of technological progress	0.1%
$\bar{b}, \bar{d}$	Birth/Death rate at subsistence	2.75%, 3.02%
φ_b, φ_d	Income elasticities (Birth, Death)	1.41, -0.55
$k_M/k_B - 1$	City excess mortality	0.5
h_M	Maximum trade and war effect	0.333
τ	Tax rate on above-subsistence income	0.52
f_W	Fixed cost of warfare (per pop)	0.011
A_W	Belligerence parameter (Europe)	22.54
A_W^{China}	Belligerence parameter (China)	5.63
ν	Upper bound on city growth	0.46%

The Transition Path (Figure 6)

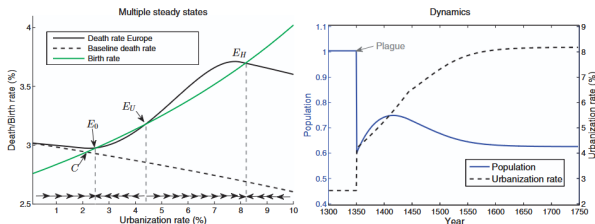


FIGURE 6

Long-run impact of the plague with “Horsemen effect”—The case of Europe

- **Pre-1348:** Europe is in E_0 .
- **1348–1350:** Population falls by 40%.
- **Post-Plague:** Wages rise → urbanization spikes → taxes for war surge → death rates stay high.
- **Outcome:** Gains become permanent; the economy converges to E_H rather than reverting to E_0 .

Why China Was Different

- **Unified Politics:** China was a consolidated empire. Higher city wealth did not translate into more frequent inter-state warfare.
- **Healthier Cities:** Due to better hygiene (transfer of “night soil”) and diet, Chinese cities were not death-traps.
- **Low $h(n_M)$:** Mortality did not rise much with income.
- **Unique Steady State:** In China, the mortality schedule crossed the birth schedule only once (at low income).
- **Effect of Plague:** Temporary gain in p.c. income followed by population recovery and a return to the original subsistence level.

Technological Progress (Figure 7)

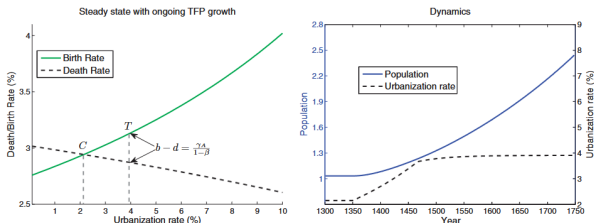
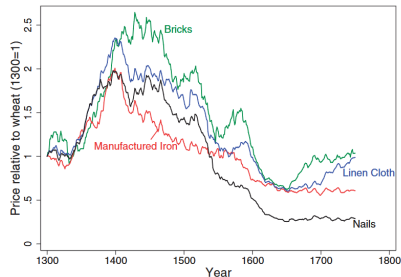
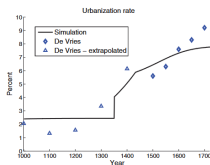
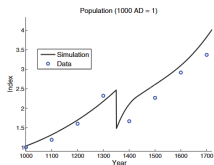


FIGURE 7
Effect of ongoing technological progress

- **Traditional View:** TFP growth drives the rise of Europe.
- **Paper's Finding:** TFP growth at 0.1% p.a. (pre-industrial norm) raises population but barely moves p.c. income.
- **Comparison:** The “Horsemen Effect” is orders of magnitude more powerful than pre-modern technological change in explaining income gains.

Empirical Verification (Figs 8 & 9)



- **Model Fit:** Simulation matches population and urbanization data from 1000 to 1700.
- **Prices:** Confirmation of Engel's Law—relative prices of urban goods (bricks, linen) peaked shortly after the plague.

- **Takeaway:** Paradoxically, death and disease spelled riches. By keeping population low, the Horsemen allowed Europe to maintain high wages.
- **European Marriage Pattern (EMP):** An additional mechanism where Europeans postponed marriage to limit births, further boosting incomes.
- **Final Insight:** The "First Divergence" was a result of unique European geographical and political conditions interacting with the Black Death shock.